Feature Report: Master Features

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Basic Information

- **Total Customers**: 61,410
- **Total Features**: 182
- **Feature Columns**: kyc_type, txn_count_total, txn_per_day_avg, txn_velocity, high_frequency_customer_flag, volume_debit_total, volume_credit_total, account_turnover_rate, flow_through_velocity_hours, time_between_deposit_withdrawal, sudden_inflow_outflow_pattern, txn_count_eft, volume_eft_total, txn_velocity_eft, volume_eft_sudden_increase, txn_count_eft_spike, velocity_change_eft, amount_mean, amount_stddev, amount_min, amount_max, amount_median, round_amount_count, round_amount_pct, round_amount_flag, just_below_threshold_count, structuring_pattern_flag, large_txn_count_10k, large_txn_count_50k, large_txn_count_100k, amount_max_ratio_mean, amount_stddev_ratio_mean, amount_cv_by_channel, amount_near_50k_increment_count, channels_used_count, multi_channel_flag, wire_txn_count, wire_volume_total, wire_volume_avg, wire_volume_max, wire_large_count, has_wire_transfers, abm_cash_txn_count, abm_cash_volume, abm_cash_volume_avg, abm_cash_volume_max, abm_cash_large_count, abm_cash_pct, structured_cash_deposits_same_day, western_union_txn_count, western_union_volume_total, western_union_volume_avg, has_western_union, eft_txn_count, eft_volume_total, eft_volume_avg, emt_txn_count, emt_volume_total, emt_volume_avg, card_ecommerce_txn_count, card_ecommerce_pct, card_txn_count, card_volume_total, card_volume_avg, cheque_txn_count, cheque_volume_total, cheque_volume_avg, cheque_volume_max, cheque_large_count, country_diversity, province_diversity, city_diversity, cross_border_txn_count, cross_border_txn_pct, cross_border_flag, is_country_financial_hub, is_country_offshore_structure_jurisdiction, is_country_trade_conduit, is_country_private_banking_hub, is_country_shell_company_jurisdiction, is_country_tbml_high_risk, is_country_major_port, is_country_real_estate_ml_risk, is_country_hnwi_concentration, is_country_high_cash_usage, txn_per_unique_location, high_geographic_dispersion, customer_country_financial_hub, customer_country_offshore_structure_jurisdiction, customer_country_trade_conduit, customer_country_private_banking_hub, customer_country_shell_company_jurisdiction, customer_country_tbml_high_risk, customer_country_major_port, customer_country_real_estate_ml_risk, customer_country_hnwi_concentration, customer_country_high_cash_usage, txn_hour_mode, after_hours_txn_pct, after_hours_flag, weekend_txn_count, weekend_txn_pct, weekend_heavy_flag, midnight_txn_count, early_morning_txn_count, late_night_txn_count, midnight_txn_pct, unusual_hours_flag, time_between_txn_mean, time_between_txn_median, time_between_txn_min, time_between_txn_max, time_between_txn_std, very_short_time_between_txn, rapid_fire_txn_count, rapid_fire_flag, txn_hour_std, consistent_timing_flag, business_hours_txn_pct, non_business_hours_heavy, months_active, txn_per_month, total_volume, avg_transaction_amount, max_transaction_amount, transaction_amount_std, amount_to_income_ratio, amount_to_sales_ratio, amount_to_revenue_ratio, lifestyle_mismatch, severe_lifestyle_mismatch, has_missing_income, has_missing_sales, amount_cv, high_amount_variability, debit_volume, credit_volume, net_flow, debit_credit_ratio, high_debit_ratio, max_txn_to_income_ratio, max_txn_to_sales_ratio, max_txn_to_revenue_ratio, single_txn_exceeds_revenue, txn_count_total_behavioral, transaction_consistency, consistent_pattern_flag, behavioral_risk_score, high_behavioral_risk, account_age, business_age, new_account_flag, very_new_business_flag, is_oil_company, is_msb_business, is_maybe_massage_parlor, is_ecom_business, is_hotel_or_motel, is_shell_company, is_cash_intensive, is_real_estate, is_precious_metal_business, is_professional_service, is_transportation, is_vague_category, has_missing_income_profile, has_missing_sales_profile, has_missing_industry_occupation, has_missing_country, has_missing_province, has_missing_city, industry_code_high_risk_trade, suspicious_occupation_flag, is_individual, is_business, profile_risk_score, high_profile_risk, txn_score_max, txn_score_mean, txn_score_std, txn_score_count_above_threshold, txn_score_pct_above_threshold

Data Quality Checks

☑ No Null Values

All features have complete data.

☑ No Duplicate Customer IDs

Feature Statistics

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero
kyc_type	object	N/A	N/A	N/A	N/A	N/A	0	0.00
txn_count_total	Integer	1	4,991	96.13	63	102.92	0	0.00
txn_per_day_avg	Float	0.00	393.00	1.19	0.72	3.97	907	1.48
txn_velocity	Float	0.00	393.00	1.19	0.72	3.97	907	1.48
high_frequency_customer_flag	Integer	0	1	0.01	0	0.08	61,045	99.4
volume_debit_total	Float	0.00	1437637276.00	2861381.83	886504.50	14983096.02	550	0.90
volume_credit_total	Float	0.00	2052481891.00	3259518.81	1213366.50	18248172.39	2,080	3.39
account_turnover_rate	Float	0.00	10.00	1.87	0.77	2.73	550	0.90
flow_through_velocity_hours	Float	0.00	2184.00	57.24	28.17	108.68	4,391	7.15
time_between_deposit_withdrawal	Float	0.00	2184.00	57.24	28.17	108.68	4,391	7.15
sudden_inflow_outflow_pattern	Integer	0	1	0.79	1	0.41	13,171	21.4
txn_count_eft	Integer	0	4,959	17.44	11	43.14	10,863	17.6
volume_eft_total	Float	0.00	2675109760.00	2750979.67	790347.50	21009517.50	10,863	17.6

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero
txn_velocity_eft	Float	0.00	53.35	0.24	0.16	0.48	10,863	17.6
volume_eft_sudden_increase	Integer	0	1	0.16	0	0.37	51,582	84.0
txn_count_eft_spike	Integer	0	0	0.00	0	0.00	61,410	100.
velocity_change_eft	Integer	0	0	0.00	0	0.00	61,410	100.
amount_mean	Float	0.00	224838493.00	98145.85	35126.97	1135873.73	6	0.01
amount_stddev	Float	0.00	483604743.40	301463.30	66953.72	2721253.14	702	1.14
amount_min	Float	0.00	224838493.00	7101.84	99.00	910656.87	28,888	47.0
amount_max	Float	0.00	2051810761.00	2248623.79	350000.00	15858254.32	6	0.01
amount_median	Float	0.00	224838493.00	24742.14	8328.75	916757.00	6	0.01
round_amount_count	Integer	0	150	1.32	0	3.10	37,115	60.4
round_amount_pct	Float	0.00	1.00	0.02	0.00	0.05	37,115	60.4
round_amount_flag	Integer	0	1	0.00	0	0.02	61,396	99.9
just_below_threshold_count	Integer	0	33	0.06	0	0.41	58,770	95.7
structuring_pattern_flag	Integer	0	1	0.00	0	0.06	61,178	99.6
large_txn_count_10k	Integer	0	320	0.64	0	3.55	48,078	78.2
large_txn_count_50k	Integer	0	56	0.12	0	0.71	57,108	92.9
large_txn_count_100k	Integer	0	26	0.05	0	0.36	59,006	96.0
amount_max_ratio_mean	Float	0.00	100.00	18.40	11.95	19.56	6	0.01
amount_stddev_ratio_mean	Float	0.00	10.00	2.45	2.09	1.54	702	1.14
amount_cv_by_channel	Float	0.00	10.00	1.52	1.41	0.69	791	1.29
amount_near_50k_increment_count	Integer	0	1,101	60.15	29	74.88	1,585	2.58
channels_used_count	Integer	1	7	3.36	4	1.32	0	0.00
multi_channel_flag	Integer	0	1	0.74	1	0.44	16,206	26.3
wire_txn_count	Integer	0	467	0.08	0	2.06	59,375	96.6
wire_volume_total	Float	0.00	2051810761.00	359996.42	0.00	12658290.86	59,375	96.6
wire_volume_avg	Float	0.00	2051810761.00	140638.16	0.00	8948653.09	59,375	96.6
wire_volume_max	Float	0.00	2051810761.00	283374.76	0.00	11138066.69	59,375	96.6
wire_large_count	Integer	0	65	0.03	0	0.43	60,464	98.4
has_wire_transfers	Integer	0	1	0.03	0	0.18	59,375	96.6
abm_cash_txn_count	Integer	0	294	2.68	0	6.06	30,984	50.4
abm_cash_volume	Float	0.00	7729793.00	78347.04	0.00	201468.45	30,984	50.4
abm_cash_volume_avg	Float	0.00	1890000.00	14642.29	0.00	27433.95	30,984	50.4
abm_cash_volume_max	Float	0.00	5473000.00	32498.58	0.00	75546.76	30,984	50.4
abm_cash_large_count	Integer	0	8	0.00	0	0.07	61,245	99.7
abm_cash_pct	Float	0.00	1.00	0.45	0.00	0.47	30,984	50.4
structured_cash_deposits_same_day	Integer	0	1	0.00	0	0.01	61,407	100.
western_union_txn_count	Integer	0	24	0.03	0	0.47	60,735	98.9
western_union_volume_total	Float	0.00	1325575.00	1244.83	0.00	18111.68	60,735	98.9
western_union_volume_avg	Float	0.00	410378.00	402.98	0.00	4859.64	60,735	98.9
has_western_union	Integer	0	1	0.01	0	0.10	60,735	98.9
eft_txn_count	Integer	0	4,959	17.44	11	43.14	10,863	17.6
eft_volume_total	Float	0.00	2675109760.00	2750979.67	790347.50	21009517.50	10,863	17.6
eft_volume_avg	Float	0.00	93885128.67	103304.91	60811.11	654431.91	10,863	17.6
emt_txn_count	Integer	0	1,948	13.78	6	31.19	18,518	30.1
emt_volume_total	Float	0.00	81206623.00	584529.65	176767.00	1618428.24	18,518	30.1

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero
emt_volume_avg	Float	0.00	1100756.50	29378.88	23057.12	39124.60	18,518	30.1
card_ecommerce_txn_count	Integer	0	155	8.47	2	13.01	21,458	34.9
card_ecommerce_pct	Float	0.00	1.00	0.12	0.12	0.13	21,458	34.9
card_txn_count	Integer	0	974	57.86	17	85.71	12,184	19.8
card_volume_total	Float	0.00	26289170.00	566823.66	120248.50	1358013.20	12,295	20.0
card_volume_avg	Float	0.00	3581565.00	8714.93	5561.91	39065.45	12,295	20.0
cheque_txn_count	Integer	0	1,186	3.92	0	19.38	31,806	51.7
cheque_volume_total	Float	0.00	867939383.00	1719608.04	0.00	13954568.59	31,806	51.7
cheque_volume_avg	Float	0.00	113648725.00	147925.21	0.00	1176734.58	31,806	51.7
cheque_volume_max	Float	0.00	567888708.00	921058.13	0.00	7782465.08	31,806	51.7
cheque_large_count	Integer	0	103	0.24	0	1.67	55,686	90.6
country_diversity	Integer	0	12	2.34	2	1.77	11,979	19.5
province_diversity	Integer	0	20	6.60	7	4.83	11,980	19.5
city_diversity	Integer	0	96	16.42	9	17.61	11,972	19.5
cross_border_txn_count	Integer	0	196	10.83	3	16.36	19,452	31.6
cross_border_txn_pct	Float	0.00	1.00	0.14	0.14	0.14	19,452	31.6
cross_border_flag	Integer	0	1	0.68	1	0.47	19,452	31.6
is_country_financial_hub	Integer	0	1	0.39	0	0.49	37,707	61.4
is_country_offshore_structure_jurisdiction	Integer	0	1	0.13	0	0.34	53,429	87.0
is_country_trade_conduit	Integer	0	1	0.08	0	0.28	56,219	91.5
is_country_private_banking_hub	Integer	0	1	0.07	0	0.26	56,958	92.7
is_country_shell_company_jurisdiction	Integer	0	1	0.19	0	0.39	49,921	81.2
is_country_tbml_high_risk	Integer	0	1	0.08	0	0.28	56,219	91.5
is_country_major_port	Integer	0	1	0.08	0	0.28	56,219	91.5
is_country_real_estate_ml_risk	Integer	0	1	0.80	1	0.40	12,360	20.1
is_country_hnwi_concentration	Integer	0	1	0.07	0	0.26	56,958	92.7
is_country_high_cash_usage	Integer	0	1	0.04	0	0.20	58,802	95.7
txn_per_unique_location	Float	0.00	15.00	1.64	1.53	1.16	11,972	19.5
high_geographic_dispersion	Integer	0	1	0.67	1	0.47	20,221	32.9
customer_country_financial_hub	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_offshore_structure_jurisdiction	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_trade_conduit	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_private_banking_hub	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_shell_company_jurisdiction	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_tbml_high_risk	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_major_port	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_real_estate_ml_risk	Integer	1	1	1.00	1	0.00	0	0.00
customer_country_hnwi_concentration	Integer	0	0	0.00	0	0.00	61,410	100.
customer_country_high_cash_usage	Integer	0	0	0.00	0	0.00	61,410	100.
txn_hour_mode	Integer	-1	23	11.99	12	4.18	992	1.62
after_hours_txn_pct	Float	0.00	1.00	0.43	0.43	0.14	1,663	2.71
after_hours_flag	Integer	0	1	0.19	0	0.39	49,817	81.1
weekend_txn_count	Integer	0	2,039	30.42	21	32.84	2,157	3.51
weekend_txn_pct	Float	0.00	1.00	0.35	0.33	0.13	2,157	3.51
weekend_heavy_flag	Integer	0	1	0.68	1	0.47	19,643	31.9

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero
midnight_txn_count	Integer	0	467	5.77	4	7.31	8,850	14.4
early_morning_txn_count	Integer	0	225	7.43	5	7.46	6,898	11.2
late_night_txn_count	Integer	0	385	6.69	4	8.29	9,038	14.7
midnight_txn_pct	Float	0.00	1.00	0.07	0.06	0.09	8,850	14.4
unusual_hours_flag	Integer	0	1	0.87	1	0.34	8,070	13.1
time_between_txn_mean	Float	0.00	2194.13	74.00	35.23	139.42	1,660	2.70
time_between_txn_median	Float	0.00	2194.13	57.91	23.38	134.76	1,660	2.70
time_between_txn_min	Float	0.00	2194.13	16.08	0.22	112.30	1,660	2.70
time_between_txn_max	Float	0.00	2194.13	230.56	162.46	226.25	1,660	2.70
time_between_txn_std	Float	0.00	1510.38	61.34	34.58	88.29	2,233	3.64
very_short_time_between_txn	Integer	0	913	1.33	0	6.21	38,323	62.4
rapid_fire_txn_count	Integer	0	1,662	9.97	2	25.46	20,785	33.8
rapid_fire_flag	Integer	0	1	0.25	0	0.43	46,191	75.2
txn_hour_std	Float	0.00	16.26	5.04	5.09	1.21	1,826	2.97
consistent_timing_flag	Integer	0	1	0.00	0	0.04	61,333	99.8
business_hours_txn_pct	Float	0.00	1.00	0.37	0.38	0.12	2,189	3.56
non_business_hours_heavy	Integer	0	1	0.06	0	0.23	57,822	94.1
months_active	Integer	0	3	2.89	3	0.47	907	1.48
txn_per_month	Float	0.00	1663.67	30.80	19.67	33.84	907	1.48
total_volume	Float	0.00	2677539042.00	6120900.64	2311184.50	28812801.85	6	0.01
avg_transaction_amount	Float	0.00	224838493.00	98145.85	35126.97	1135873.73	6	0.01
max_transaction_amount	Float	0.00	2051810761.00	2248623.79	350000.00	15858254.32	6	0.01
transaction_amount_std	Float	0.00	483604743.40	301463.30	66953.72	2721253.14	702	1.14
amount_to_income_ratio	Float	0.00	641237.23	266.81	0.22	3365.03	22,992	37.4
amount_to_sales_ratio	Float	0.00	453407871.00	58007.52	0.00	3101551.92	57,379	93.4
amount_to_revenue_ratio	Float	0.00	453407871.00	58274.34	0.35	3101548.75	18,961	30.8
lifestyle_mismatch	Integer	0	1	0.22	0	0.41	48,063	78.2
severe_lifestyle_mismatch	Integer	0	1	0.17	0	0.37	51,072	83.1
has_missing_income	Integer	0	1	0.24	0	0.43	46,732	76.1
has_missing_sales	Integer	0	1	0.07	0	0.25	57,132	93.0
amount_cv	Float	0.00	10.00	2.45	2.09	1.54	702	1.14
high_amount_variability	Integer	0	1	0.54	1	0.50	28,297	46.0
debit_volume	Float	0.00	1437637276.00	2861381.83	886504.50	14983096.02	550	0.90
credit_volume	Float	0.00	2052481891.00	3259518.81	1213366.50	18248172.39	2,080	3.39
net_flow	Float	-628883171.00	2052309612.00	398136.98	191413.50	16876029.55	6	0.01
debit_credit_ratio	Float	0.00	10.00	1.54	0.70	2.29	2,624	4.27
high_debit_ratio	Integer	0	1	0.19	0	0.39	49,627	80.8
max_txn_to_income_ratio	Float	0.00	630987.45	98.18	0.04	3157.55	22,992	37.4
max_txn_to_sales_ratio	Float	0.00	299434000.00	23841.46	0.00	1512651.10	57,379	93.4
max_txn_to_revenue_ratio	Float	0.00	299434000.00	23939.64	0.06	1512652.85	18,961	30.8
single_txn_exceeds_revenue	Integer	0	1	0.19	0	0.39	49,939	81.3
txn_count_total_behavioral	Integer	1	4,991	96.13	63	102.92	0	0.00
transaction_consistency	Float	0.00	1.00	0.33	0.32	0.12	6	0.01
consistent_pattern_flag	Integer	0	0	0.00	0	0.00	61,410	100.
behavioral_risk_score	Integer	0	9	2.35	1	2.66	12,844	20.9

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero
high_behavioral_risk	Integer	0	1	0.25	0	0.44	45,809	74.6
account_age	Integer	0	123	14.01	11	11.68	4,862	7.92
business_age	Integer	0	155	40.24	39	23.78	3,936	6.41
new_account_flag	Integer	0	1	0.08	0	0.27	56,548	92.0
very_new_business_flag	Integer	0	1	0.01	0	0.08	61,000	99.3
is_oil_company	Integer	0	1	0.00	0	0.03	61,340	99.8
is_msb_business	Integer	0	1	0.01	0	0.11	60,723	98.8
is_maybe_massage_parlor	Integer	0	1	0.00	0	0.04	61,293	99.8
is_ecom_business	Integer	0	1	0.01	0	0.07	61,085	99.4
is_hotel_or_motel	Integer	0	1	0.00	0	0.02	61,374	99.9
is_shell_company	Integer	0	1	0.02	0	0.15	60,081	97.8
is_cash_intensive	Integer	0	1	0.00	0	0.07	61,134	99.5
is_real_estate	Integer	0	1	0.00	0	0.07	61,147	99.5
is_precious_metal_business	Integer	0	1	0.00	0	0.03	61,366	99.9
is_professional_service	Integer	0	1	0.01	0	0.11	60,697	98.8
is_transportation	Integer	0	1	0.01	0	0.08	61,015	99.3
is_vague_category	Integer	0	1	0.01	0	0.08	61,048	99.4
has_missing_income_profile	Integer	0	1	0.37	0	0.48	38,421	62.5
has_missing_sales_profile	Integer	0	1	0.93	1	0.25	4,033	6.57
has_missing_industry_occupation	Integer	0	1	0.00	0	0.04	61,303	99.8
has_missing_country	Integer	0	0	0.00	0	0.00	61,410	100.
has_missing_province	Integer	0	0	0.00	0	0.00	61,410	100.
has_missing_city	Integer	0	0	0.00	0	0.00	61,410	100.
industry_code_high_risk_trade	Integer	0	1	0.02	0	0.15	60,068	97.8
suspicious_occupation_flag	Integer	0	1	0.02	0	0.15	59,906	97.5
is_individual	Integer	0	1	0.86	1	0.34	8,311	13.5
is_business	Integer	0	1	0.14	0	0.34	53,099	86.4
profile_risk_score	Integer	1	13	1.58	1	1.20	0	0.00
high_profile_risk	Integer	0	1	0.07	0	0.26	57,099	92.9
txn_score_max	Float	0.00	0.60	0.13	0.20	0.12	22,366	36.4
txn_score_mean	Float	0.00	0.60	0.01	0.00	0.02	22,366	36.4
txn_score_std	Float	0.00	0.42	0.03	0.02	0.03	22,440	36.5
txn_score_count_above_threshold	Integer	0	26	0.01	0	0.20	60,978	99.3
txn_score_pct_above_threshold	Float	0.00	1.00	0.00	0.00	0.01	60,978	99.3

Validation Checks

- [x] No Unexpected Negative Values
- [x] No Infinite Values

Outlier Detection (Values > 3 Standard Deviations)

- `txn_count_total`: 924 outliers (1.50%)
- `txn_per_day_avg`: 145 outliers (0.24%)
- `txn_velocity`: 145 outliers (0.24%)
- `high_frequency_customer_flag`: 365 outliers (0.59%)
- `volume_debit_total`: 544 outliers (0.89%)
- `volume_credit_total`: 363 outliers (0.59%)
- `flow_through_velocity_hours`: 1,009 outliers (1.64%)
- `time_between_deposit_withdrawal`: 1,009 outliers (1.64%)

- `txn_count_eft`: 380 outliers (0.62%)
- `volume_eft_total`: 326 outliers (0.53%)

... and 127 more features with outliers

Flag Feature Distributions

Feature	True Count	True Percentage	False Count	False Percentage
`high_frequency_customer_flag`	365	0.59%	61,045	99.41%
`round_amount_flag`	14	0.02%	61,396	99.98%
`structuring_pattern_flag`	232	0.38%	61,178	99.62%
`multi_channel_flag`	45,204	73.61%	16,206	26.39%
`cross_border_flag`	41,958	68.32%	19,452	31.68%
`after_hours_flag`	11,593	18.88%	49,817	81.12%
`weekend_heavy_flag`	41,767	68.01%	19,643	31.99%
`unusual_hours_flag`	53,340	86.86%	8,070	13.14%
`rapid_fire_flag`	15,219	24.78%	46,191	75.22%
`consistent_timing_flag`	77	0.13%	61,333	99.87%
`consistent_pattern_flag`	0	0.00%	61,410	100.00%
`new_account_flag`	4,862	7.92%	56,548	92.08%
`very_new_business_flag`	410	0.67%	61,000	99.33%
`suspicious_occupation_flag`	1,504	2.45%	59,906	97.55%

Summary

- ☑ **Total Features**: 182
- ☑ **Data Completeness**: 0.55%
- ☑ **Customers Covered**: 61,410
- ☑ **Flag Features**: 14

Report generated automatically by feature engineering pipeline